

Feifan Liu

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EDUCATION

University of Michigan – Ann Arbor

Master of Science in Information – Big Data Analytics Track, GPA: 4.0/4.0

Ann Arbor, Michigan, USA

Aug. 2025 – Jun. 2027

University of Toronto

Bachelor of Science: Specialist in Statistical Science: Method and Practice

Toronto, Ontario, Canada

Sep. 2021 – Nov. 2024

Bachelor of Science: Major in Economics: Focus in Data Analytics

Sep. 2021 – Nov. 2024

- Honors B.S. with High Distinction, Dean's List Scholar in the Faculty of Art & Science, GPA: 3.9/4.0
- Summary:** Completed a specialist & a major one year ahead of schedule, including the most challenging and selective Specialist in Statistics and a major in Economics

Work Experience

Allcorehattrass Technology Co., Ltd

Nanjing, Jiangsu, China

Machine Learning Engineer Intern

Nov. 2024 – Apr. 2025

- Utilized graph classification, deep learning, and convolutional neural networks (CNN) to help drone identify various types of pins, increasing identification accuracy from 25% to 90%.
- Drone Route Optimization and Automatic Task Allocation: Explored and customized DB-GPT's task scheduling and decision-making capabilities by evaluating and deploying various LLMs (e.g., Qwen-1.8B, InternLM-7B/20B, Qwen-Max/Turbo APIs), utilizing deployment strategies such as local installations, Docker environments, and API-based integration for large-scale models (32B parameters).
- Developed and enhanced a data assistant app using AWEL (Agentic Workflow Expression Language), integrating tools for intelligent Q&A across databases, knowledge bases, and internet sources with real-time retrieval from verified websites, and designed workflows to streamline tower model identification and defect detection model integration.

Shenwan Hongyuan Asset Management

Shanghai, China

Equity & Multi-Investment Intern

Jun. 2024 – Aug. 2024

- Participated in managing a RMB 50 million Fund of Funds (FoF) by preparing weekly and monthly performance reports with prescriptive analytics, attributing results to market factors, and creating Tableau dashboards for data visualization.
- Contributed to the strategic adjustment of the Barra Multi-Factor model by collecting and analyzing a dataset of 2,000 stocks over one year; performed descriptive analytics and calculated excess returns using Python, and utilized OLS regression and t-tests to determine model parameters.
- Implemented a customized Snowball pricing model, which depends on the price volatility of the underlying product (CSI 500 Index), and determined the number of knock-outs and knock-ins at +/- 3%

Deloitte

Shanghai, China

Cyber Risk Advisory & Data Analysis Intern

May. 2023 – Aug. 2023

- Conducted cybersecurity research on system localization by designing a questionnaire, collecting 400 responses, and performing quantitative and qualitative analysis with statistical visualizations in Python; presented findings to relevant stakeholders.
- Analyzed the internal business structure of client companies (retail industry), to understand tasks processing methods and adopted software systems of various departments (e.g., business operation, supply chain)

BlackFlame Medical

Shanghai, China

Data Analyst Intern

July. 2021 – Nov. 2021

- Data analysis for clinical diagnosis, especially in Scoliosis correction: extracted and processed data from medical database, fit the relationship between patients' characteristics (Scoliosis angle, number of damaged vertebrae, pain level, etc.) and surgical types with Python, to assist in clinical diagnosis

PROJECTS

<https://github.com/Aaquilaaaaa/Projects.git>

PRISM(Process Reasoning & Interpretable System for Manufacturing)

Ann Arbor, Michigan, USA

- Lead a team to Engineer a Physics-Informed Neural Network (PINN) equipped with an RNN encoder to predict milling chatter, overcoming the "black-box" limitations of traditional data-driven models and enhancing physical interpretability in precision manufacturing.
- Designed an unsupervised time-series labeling pipeline by extracting the coefficient of variation from steady-state force extrema and applying Gaussian Mixture Models (GMM), mitigating label noise from theoretical SLDs to achieve a 95.95% empirical classification accuracy.

Relative Belief (A+)

Toronto, Ontario, Canada

- Researched on the feasibility of automating experimental bias evaluations within probabilistic programming frameworks, applied both analytical and simulation approaches to assess hypothesis testing and estimate bias in various statistical models, including beta-binomial and linear regression frameworks
- Implemented the bias in favor and bias against calculation for different models, with R/Python; optimized the algorithm time complexity by mapping arrays to matrices to handle large-scale computations

SKILLS

Python (Pandas, NumPy, SciPy, Matplotlib, PyMC, Pytorch,...)|SQL|GitHub|C|R|Docker|Bash|Tableau